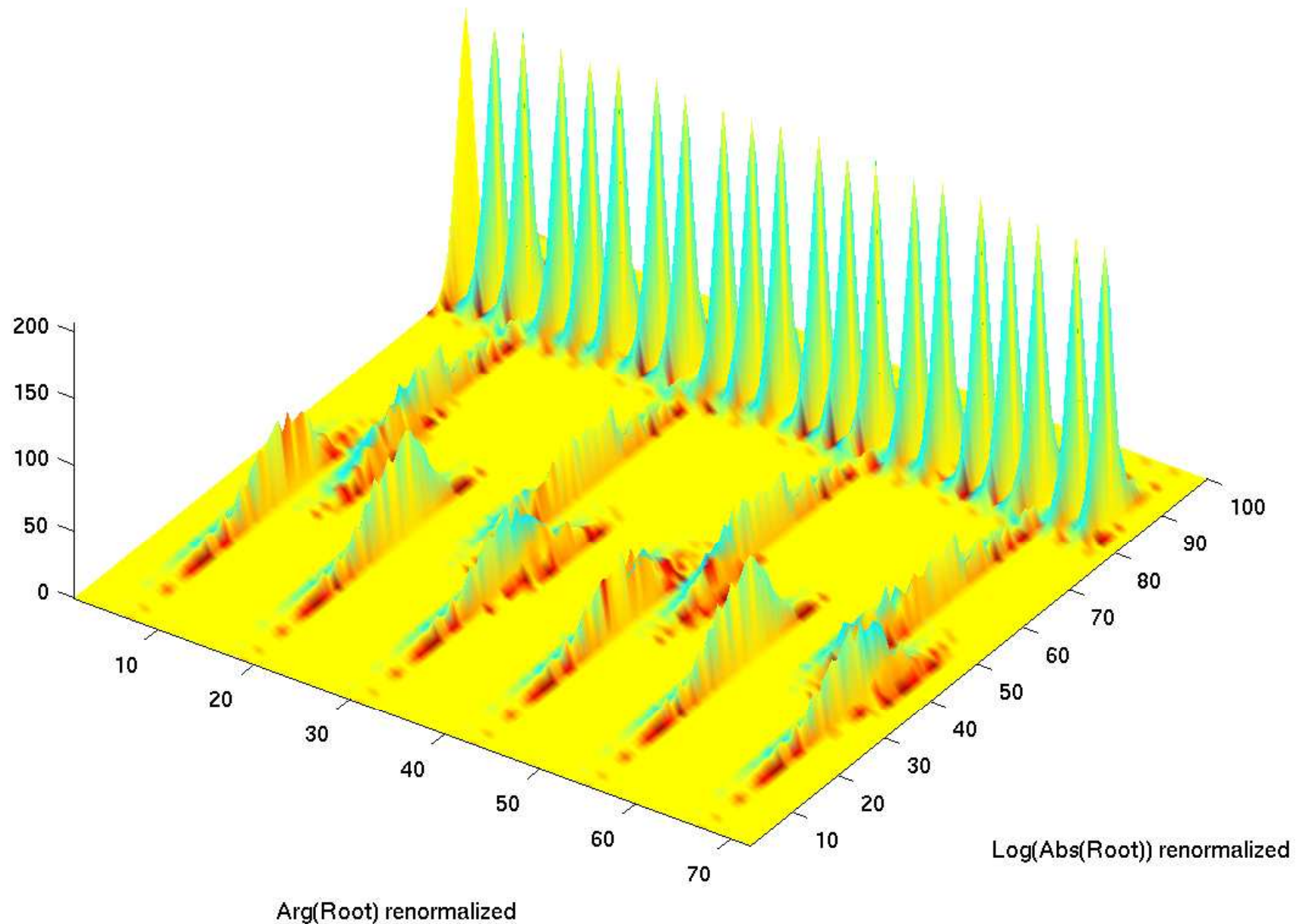


$\mathbb{R}$ AND $\mathbb{C}$ UNDER ONE ROOF?

Rescaled Histogram of $\text{Sqrt}(-1) \cdot \text{Log}(\text{Roots})$ for 1000 Random Tetranomials with Spectrum $[0,6,10,31]$



$\mathbb{R}$ AND $\mathbb{C}$ UNDER ONE ROOF?

Voorhoeve-Wilder Theorem [1976, 1912] Let $S \subseteq \mathbb{C}^*$ be any angular sector of width θ , f any univariate m -sparse real polynomial with support A , and define $\mathcal{N}(f, S)$ to be the number of roots f in S_θ . Then

$$\left| \mathcal{N}(f, S) - \frac{\theta \text{Length}(\text{Conv} A)}{2\pi} \right| \leq (m-1)\varepsilon_\theta,$$

where ε_θ is 0 or 1 according as θ is 2π or not. ■

Curious Fact: There is a **much** sharper arithmetic analogue of fewnomial theory over $\mathbb{R}$ and $\mathbb{C}$ (involving $\mathbb{Q}_p$ and $\mathbb{C}_p$), and the proofs go opposite Khovanski's approach...



Thank you for listening!

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